2024 2nd International Conference on Computer, Vision and Intelligent Technology (ICCVIT) | 979-8-3315-4004-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCVIT63928.2024.10872583

Hyperspectral and Multispectral image Fusion based on diffusion model and MergeCNN

1st Wei Li
Automation School
Beijing Information Science and Technology University
Beijing, China
2022020446@bistu.edu.cn

2nd Lu Li*
Automation School
Beijing Information Science and Technology University
Beijing, China
20192380@bistu.edu.cn

3rd Man Peng
Aerospace Information Research institute
Chinese Academy of Sciences
Beijing, China
pengman@radi.ac.cn

4th Zongzheng Yan
Automation School
Beijing Information Science and Technology University
Beijing, China
2022020448@bistu.edu.cn

***Abstract*—In recent years, diffusion models have attracted attention for Hyperspectral and Multi-spectral image fusion or Pansharping. However, a little noise often remains in the images generated by diffusion models, which affects the final quality. To mitigate this issue, we propose a diffusion model-based method for hyperspectral and multispectral image fusion. We introduce the MergeCNN module, which excels in local feature extraction and feature representation through deep convolution, and has a significant advantage in noise suppression.The MergeCNN module combined with the diffusion model further optimises the details of the image and improves the quality of the image. Experimental results on two public datasets show that DIFF-Merge exhibits advanced performance compared to existing commonly used methods.**

***Index Terms*—image fusion, diffusion models, Hyperspectral images, multispectral images.**

I. Introduction

Hyperspectral and multispectral images are two types of images widely used in remote sensing. Hyperspectral images are rich in spectral information but usually have relatively low spatial resolution. Multispectral images, while having relatively less spectral information, boast high spatial resolution. The fusion of hyperspectral and multispectral images can integrate their advantages, improving the spatial resolution of hyperspectral images and expanding their application range.

Image fusion technology processes and combines information from different imaging sensors capturing the same object or scene. It not only deepens the use of information but also optimizes image quality. In recent years, with the increasing demand for high-resolution hyperspectral images across various fields, numerous algorithms for pansharpening and the fusion of hyperspectral and multispectral images have been proposed. These algorithms can be divided into two main groups: traditional and deep learning methods. The former are fast and straightforward but may cause spectral or spatial distortion. Deep learning methods, with their powerful feature processing capabilities, have become a mainstream technique for multi-source remote sensing image fusion. Over the past few years, deep learning techniques have become prevalent in image fusion applications, and the main methods include techniques based on Convolutional Neural Network (CNN) [1]–[6], Auto-Encoder (AE) [7], Transformer Deep Network [8], and Generative Adversarial Networks (GAN) [9].

CNN-based methods excel at efficient local feature extraction but are limited by boundary effects and spatial insensitivity when addressing global dependencies [4], [5]. In contrast, Transformer techniques excel at capturing long-range dependencies and are often used in conjunction with CNNs to enhance feature complementarity [8]. Auto-Encoders achieve image fusion through an encoding-decoding structure, capable of handling multi-scale features but deficient in preserving boundary information. GAN methods demonstrate strong potential in remote sensing image fusion but require further optimization due to their complex training process and susceptibility to the pattern collapse problem [10].

In recent years, diffusion models have been widely recognised for their stable training process and excellent image generation capabilities. This model performs well in generative tasks and is widely used in conditional or unconditional generation. This includes audio generation, text-to-image generation and other image processing tasks [11]. Diffusion models create samples by gradually sampling from a Markov chain, beginning with a recognizable simple distribution and eventually reaching a target data distribution. The advantages are the stability of the training process, minimal risk of modal collapse, and the flexibility of being able to train with only the loss of mean square error.

Recently, methods such as DDIF [10] have introduced denoising diffusion probabilistic models (DDPM) to tackle the image fusion task. These methods generate fused images by simulating the diffusion process of data in the latent space, gradually transforming the initial distribution into the target distribution. They can effectively enhance spatial resolution

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and maintain spectral accuracy. However, the images generated by diffusion models often retain a little noise.

To mitigate this issue, we propose a fusion network, DIFF-Merge, to further enhance the fusion effect and make the final result smoother and more accurate. Through deep fusion, the clarity and stability of the generated images are improved, especially excelling in detail and edge recovery. Experimental results on two public datasets show that DIFF-Merge exhibits advanced performance in multispectral and hyperspectral image fusion compared to existing commonly used methods.

II. Main Method

A. Overview of the model

We construct a diffusion model-based hyperspectral and multispectral fusion network designed to enhance the spatial and spectral resolution of images. The network combines the detailed spectral information from hyperspectral images with the sharp spatial details from multispectral images. Considering that CNNs can extract local features more efficiently through their deep convolution and achieve robust noise suppression through layer-by-layer feature representation, and have significant advantages in dealing with subtle noise. We introduce a MergeCNN module to mitigate the issue of generating images with a little remaining noise using a diffusion model, further optimizing the fusion process.

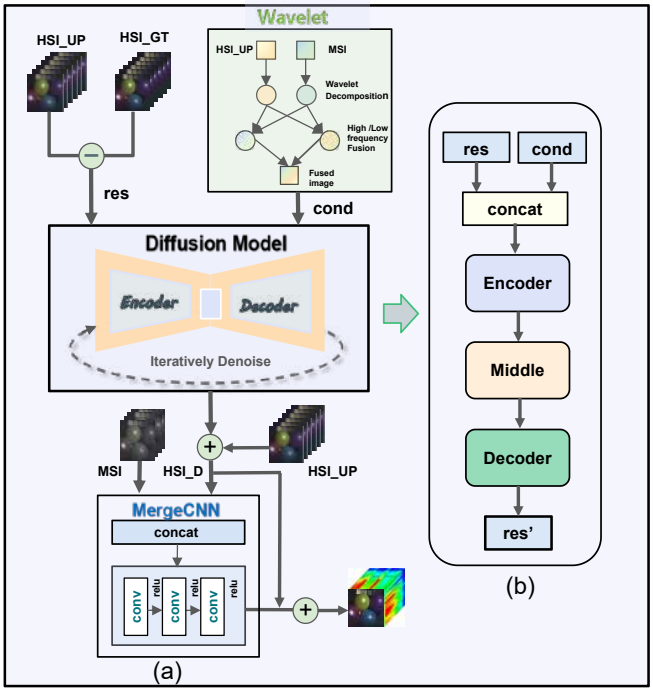


Fig. 1. The DIFF-Merge flowchart.

The model includes a DDPM-based fusion module and a MergeCNN fusion module. Figure 1(a) illustrates the overall flow. To reduce computational complexity, we take the result obtained by differencing the hyperspectral image (HSI) with its up-sampled version (HSI_UP) as the input to the diffusion model, while the MSI and HSI_UP fused by the wavelet transform are used as conditions. The diffusion model adopts the U-Net structure, and we use it to generate the difference map between HSI and HSI_UP, which we sum with HSI_UP to obtain HSI_D. Subsequently, the MSI and HSI_D are combined and then fused by a three-layer MergeCNN, and the result is summed with HSI_D to obtain the final high-resolution hyperspectral image (HRHS).

B. Diffusion module

The diffusion model primarily consists of two stages: the initia noise addition and backward denoising. The forward noise addition process is the process of gradually transforming the real data into noise. Setting the real data in the dataset as x_0, we form a series of data with noise x_t by adding noise at each time step t. The process can be expressed by the following equation:

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t}\epsilon \tag{1}$$

where $\epsilon \sim \mathcal{N}(0, I)$ is the noise sampled from the standard normal distribution. α_t is a parameter defined by the noise plan, usually taking values between ([0, 1]), reflecting the intensity of the noise. It is generally defined as: $\alpha_t = 1 - \beta_t$ where β_t is the variance of the noise added at time step t, usually linear or cosine decay: $\beta_t = \text{schedule}(t)$

The inverse denoising process starts from the noise samples x_T and removes the noise step by step to recover the original data.

During training, our target is to make the network able to accurately predict the noise at a given time step t. The loss function is usually a mean square error (MSE) loss of the form:

$$L(\theta) = \mathbb{E}_{t,x_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right] \tag{2}$$

and the loss function is optimized by comparing the noise predicted by the model with the true noise.

At the beginning of the generation phase, initial noise samples x_T are sampled from a standard normal distribution:

$$x_T \sim \mathcal{N}(0, I) \tag{3}$$

Subsequently, the inverse denoising process is carried out iteratively to gradually generate high-quality samples until it reaches x_0. The iterative formula for the process is:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_{t-1}}}\left(x_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{1-\alpha_T}}\epsilon_\theta(x_t, t)\right) \tag{4}$$

where $\epsilon_\theta(x_t, t)$ is a neural network model that is responsible for predicting the noise at a time step t.

Our diffusion model architecture, based on U-Net, is composed of three components: the encoder, the middle layer, and the decoder. Both the decoder and encoder have a three-layer structure, whereas the middle layer has a two-layer structure. The structure of all layers is the same and its detailed structure is shown in Figure 2. The model comprises a self-attention layer, GroupNorm, SiLU activation function, Dropout, and Conv2d layers. The self-attention layer highlights key information, aiding in capturing subtle features

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and enhancing generation quality. GroupNorm standardizes the data and facilitates model convergence. The SiLU activation function fits the linear relationship of the data better than a traditional sigmoid, which speeds up learning and improves stability, while the Dropout prevents overfitting and enhances the generalisation of the model, and the Conv2d layer adapts the data to the desired dimensions. Residual connections are used between the encoder, middle layer and decoder to prevent gradient vanishing.

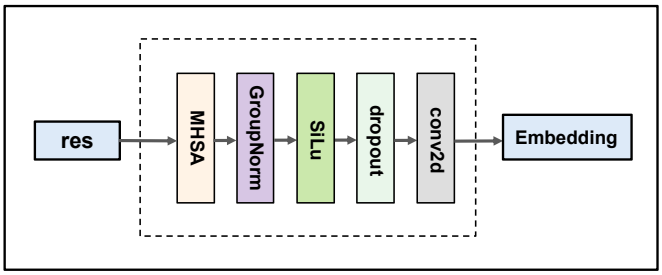


Fig. 2. The details of diffusion module.

C. MergeCNN module

The structure of MergeCNN is shown in Fig. 1(a), it consists of a concatenation unit and three convolutional layers, all equipped with a kernel size of 3, stride set to 1, and padding set to 1, and an activation function of ReLU. Since the image generated by the diffusion model may have a little remaining noise, it can be optimized through multiple layers of convolution and iteration. Specifically, the images generated by the diffusion model are initially processed and then concat operation is performed with the MSI. The result is used as the input of MergeCNN. Through the three-layer convolution operation of MergeCNN, the residual noise can be gradually filtered out, and at the same time, the information of MSI and HSI_UP can be reintegrated to further improve the quality of the fused image. The MergeCNN can effectively enhance image details and clarity, ensuring that the output image has better detail and less noise.

D. training process and inference process

Our model training is divided into two parts, first training the diffusion model and then training the MergeCNN module. In each round, the diffusion model is trained first, and after backpropagation, the obtained results are passed into the MergeCNN module for training, and the two parts are not uniformly backpropagated.

The training process of the diffusion model module mainly contains two processes, adding noise and denoising. When adding noise, the res is gradually added to the cosine noise step by step until a Gaussian distribution is reached. Compared with the traditional linear noise, the smoothness of cosine noise reduces the possibility of abrupt changes and better preserves the image details. In addition, cosine noise provides a smooth transition in the time dimension, which helps to improve the model's responsiveness to subtle changes. For the denoising process, multiple iterations of denoising are performed on the noisy image at each time step until it is reduced to res.

In summary, the process of adding noise converts the data distribution into Gaussian-distributed random noise. On the contrary, the denoising process seeks to learn to eliminate the random noise that was added previously. The training of the diffusion model maximizes its Variational Lower Bound (VLB) using a straightforward supervised loss function as eq. 2. For the training of the MergeCNN module, we use the mean squared error (MSE) as the loss function:

$$L_{Merge} = \mathbb{E}[(MergeCNN(HSI_D, MSI) - GT)^2] \quad (5)$$

For the inference process, we aim to learn a residual (res) suitable for a specific HSI_UP from random noise. Random noise is used as input, with the wavelet-transformed fusion result of the multispectral image (MSI) and HSI_UP serving as the condition for the diffusion model. By applying the trained diffusion model, the random noise is denoised to obtain the residul, which is then added to HSI_UP to generate HSI_D. Subsequently, HSI_UP and MSI are concatenated and fed into the trained MergeCNN, ultimately outputting a high spatial resolution hyperspectral image.

III. Experiments and Results Discussion

A. Datasets Overview

We perform quantitative and qualitative analyses on two public datasets, the CAVE and Pavia datasets, to evaluate the proposed framework. The CAVE dataset was acquired by an Apogee Alta U260 camera and contains 32 hyperspectral images and 31 multispectral images using the red, green, and blue (RGB) channels, with an image resolution of 512 × 512 pixels. The dataset covers 31 spectral channels with wavelengths ranging from 400 nm to 700 nm, with 10 nm intervals between each band. We select 20 of these images as the training set and the remaining 12 images are used as the test set. The Pavia dataset was collected by the Reflectance Optical System Imaging Spectrometer (ROSIS) sensor. It has 102 spectral bands and an image resolution of 512 × 512 pixels. We designated the upper portion of the dataset's captured areas for training purposes, while the rest served as the testing set. This testing set is comprised of four separate 256 × 256 hyperspectral patches without any overlap. For both of the above datasets, in order to reduce the computational cost, we cropped them into images with a resolution of 64 × 64 pixels for training and testing.

B. Experiments Setup and Performance metrics

The method we propose is based on Python 3.9.19 and PyTorch 2.2.2, optimizing the loss function on a Linux OS powered by a pair of NVIDIA GeForce RTX 3090 graphics cards. The diffusion model uses the AdamW optimizer with a learning rate set to 10^{-4}, and the encoder and decoder have 3 layers. For the CAVE dataset, the encoder's starting channel count is 32, whereas for Pavia, it's 102. After each encoder/decoder layer, multiplication or division adjustments are made using the factors 1, 2, and 4, which corresponded

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to the sequences of the 3 encoder/decoder layers. Instead of linear noise, we use cosine noise in the diffusion process. The MergeCNN module consists of a three-layer convolutional network using the ReLU activation function, which also employs the AdamW optimiser with a learning rate of 10^{-4} , trained with the diffusion model output as its input.

We use PSNR, ERGAS, SAM and SSIM as evaluation metrics. SAM quantifies spectral distortion, with values nearer to 0 indicating better performance; ERGAS measures the overall error, the smaller the value the better; PSNR measures the spatial quality, the higher the value the better; SSIM measures the structural similarity, with values closer to 1 indicating higher quality.

C. Comparative Experiments and Discussion

We compare the proposed method with six commonly used methods to verify the performance of DIFF-Merge. The compared methods include PNN [1], DiCNN [2], MSDCNN [3], FusionNet [4], DRPNN [5], and PanNet [6]. For a fair comparison, the relevant hyperparameters are chosen to be consistent with the original paper. We first performed experiments on the Cave dataset, and in the qualitative evaluation (As shown in Fig 3), we present fusion effects and error plots for visual analysis. Compared to the benchmark method, our results are closer to the real image (GT) in terms of colours, details and edges, and perform better in terms of detail recovery and visual effects. In the error map, the darker the blue colour and the lighter the red colour, the closer to the real value. In the quantitative evaluation, as shown in Table I, DIFF-Merge outperforms the other methods on three of the four metrics and has the second best results in terms of ERGAS values (Bolded are the best results, underlined are the second-best results). It is not difficult to find that our method has a very noticeable improvement over the second-best method in both PSNR and SSIM metrics.

TABLE I
RESULTS ON CAVE DATASET. THE OPTIMAL VALUES ARE PRESENTED IN BOLD, WHEREAS THE NEXT BEST ARE INDICATED WITH UNDERLINES.

	SAM	ERGAS	PSNR	SSIM
PanNet	3.1709	2.8659	40.1297	0.9472
MSDCNN	2.9897	2.8949	39.9050	0.9455
DiCNN	2.8914	2.5000	41.0076	0.9539
PNN	3.4741	3.1986	38.7736	0.9330
FusionNet	3.1401	2.7181	40.2041	0.9464
DRPNN	2.8488	**2.3889**	41.5133	0.9554
DIFF-Merge	**2.8185**	2.4411	**52.9659**	**0.9957**

We also conducted experiments on the Pavia dataset to evaluate model effectiveness. As shown in Fig. 4, we selected an image from the test set to demonstrate the fusion effect. Our method is closer to the real image (GT) in terms of colour, details and edges. The error map shows that our method has darker blue and less red, indicating that the results are closer to GT. The numerical evaluation in Table II shows that our model has a 15.12% and 3.11% improvement in effect in PSNR and SSIM metrics, respectively, compared to the second best method. Especially in PSNR, it is significantly higher than other methods, which indicates that DIFF-Merge greatly reduces the noise of the reconstructed image during the fusion process.

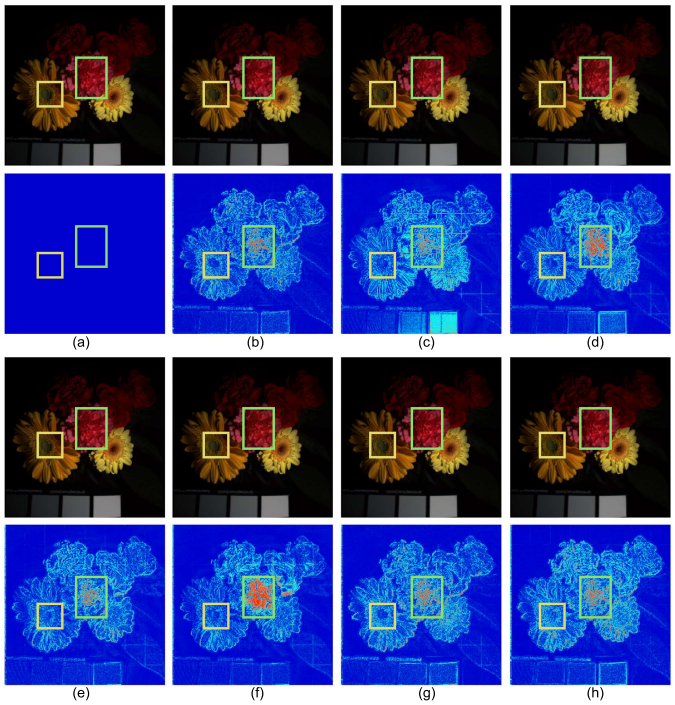


Fig. 3. The first and third rows show the true colour representation of the fusion results for our method as well as for the six compared methods, respectively. Rows two and four display the residuals between the various methods' fusion outcomes and the GT. (a) GT. (b) DIFF-Merge. (c) PNN. (d) MSDCNN. (e) FusionNet. (f) PanNet. (g) DRPNN. (h) DiCNN.

TABLE II
RESULTS ON PAVIA DATASET. THE OPTIMAL VALUES ARE PRESENTED IN BOLD, WHEREAS THE NEXT BEST ARE INDICATED WITH UNDERLINES.

	SAM	ERGAS	PSNR	SSIM
PanNet	**4.1512**	**3.2769**	30.5180	0.8853
MSDCNN	4.2562	3.4802	29.9718	0.8737
DiCNN	4.6155	3.9357	28.8406	0.8487
PNN	4.5482	3.7328	29.4047	0.8584
FusionNet	4.3758	3.5976	29.6922	0.8685
DRPNN	5.6628	4.7420	27.0552	0.8012
DIFF-Merge	4.2846	3.7217	**35.1335**	**0.9129**

D. Ablation Study

In order to test the effectiveness of the MergeCNN module, we tested the performance of the algorithm with the MergeCNN module removed and used it (DDIF) as a benchmark. We show the performance of DIFF-Merge on the cave dataset in Table III, and we can see that the MergeCNN module plays an active role. The experimental results show that compared to DDIF, DIFF-Merge achieves an improvement in the results of three metrics, and one metric is consistent with DDIF, this result verifies the effectiveness of the MergeCNN module.

In addition, we verify the necessity of inputting the residuals between HSI_GT and HSI_UP into the diffusion model. In

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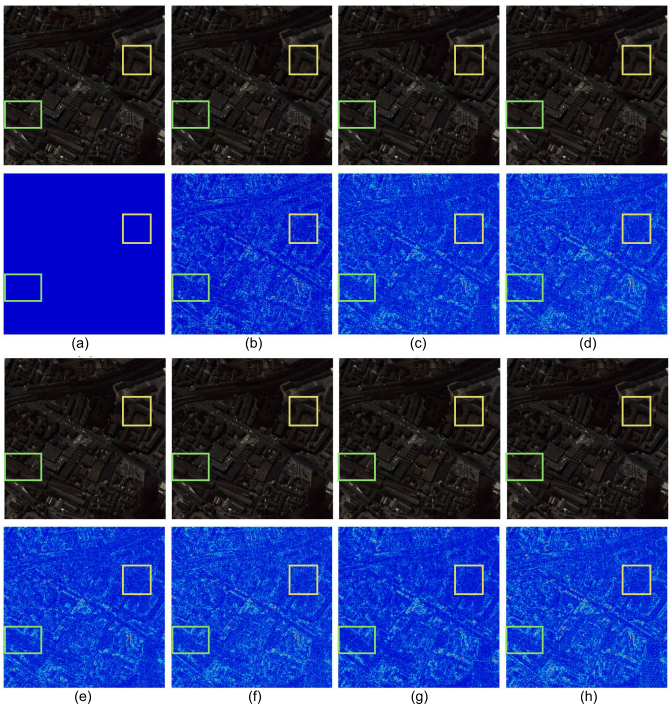


Fig. 4. The first and third rows show the true colour representation of the fusion results for our method as well as for the six compared methods, respectively. Rows two and four display the residuals between the various methods' fusion outcomes and the GT. (a) GT. (b) DIFF-Merge. (c) PNN. (d) MSDCNN. (e) FusionNet. (f) PanNet. (g) DRPNN. (h) DiCNN.

TABLE III
THE RESULTS ON ABLATION STUDY.

	SAM	ERGAS	PSNR	SSIM
DDIF	2.8598	2.4864	52.8545	**0.9957**
DIFF(HSI_GT)	8.1141	21.8114	43.6519	0.9756
DIFF-Merge	**2.8185**	**2.4411**	**52.9659**	**0.9957**

Table III, DIFF (HSI_GT) shows the results of using HSI_GT instead of residuals as input. The experimental results show that DIFF (HSI_GT) shows a significant decrease in performance on all evaluation metrics compared to DIFF-Merge. This may be due to the fact that the method consumes a large amount of computational resources, making it difficult to obtain more accurate results. At the same time, it shows that learning the residuals between HSI_UP and HSI_GT is useful for us to better perform the image fusion task.

IV. CONCLUSION

We propose a diffusion model-based hyperspectral and multispectral image fusion method named DIFF-Merge. DIFF-Merge introduces the MergeCNN module, which demonstrates excellent capability in recovering image details and edge information. Through deep feature fusion, the MergeCNN module effectively mitigates the problem of remaining noise in the diffusion model, improves the model's stability, and further optimizes the results generated by the diffusion process. We conducted comprehensive experiments on publicly available datasets such as CAVE and Pavia, and the results show that DIFF-Merge performs excellently in preserving the spatial structure and the spectral resolution of the image simultaneously. In future research, we aim to develop a KAN-based module as a bootstrap network for the diffusion model. This will serve to further strengthen the signal extraction capability and improve the generation quality and robustness of the model.

V. ACKNOWLEDGMENT

This work was supported in part by the National Natural Science Foundation of China under Grant 62471049, Open Fund of the State Key Laboratory of Remote Sensing Science under Grant OFSLRSS202317, and the Young Backbone Teacher Support Plan of Beijing Information Science and Technology University (BISTU) under Grant YBT202413.

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